

Investigating Security Issues in 5G and 4G Networks

COM-304: Communications Project Final Report

Vlad Dancau¹, Robin Bochatay², David Gu Song¹, and Quentin Marguet²

¹Department of Computer Science, EPFL

²Department of Communications Systems, EPFL

June 14, 2024

Abstract—This project investigates the security of private 4G/5G networks, focusing on the connection between the Radio Access Network (RAN) and User Equipment (UE). Using the OpenRAN standard, we simulate real-world network attacks in a controlled environment. We set up a private 5G Non-Standalone (NSA) network with open-source software and commercial hardware. Our research explores two attacks: IMSI catching and inter-generation downgrade. The IMSI catching attack intercepts unique UE identifiers via a fake base station, while the downgrade attack forces a UE to switch from a legitimate voice base station when VoLTE/VoNR is not available (usually 3G UMTS), to a spoofed 2G GSM base station, where outbound calls and SMS can be eavesdropped.

We successfully demonstrated these attacks, revealing potential security vulnerabilities. We recommend countermeasures such as RAN integrity verification and mandatory VoLTE/VoNR use to enhance 5G network security.

I. MOTIVATION

From browsing the web to critical emergency services, high speed wireless communication is an essential part of our daily lives. The fifth generation of mobile networks (5G) is the latest standard for cellular networks. It promises to bring faster speeds, lower latency, and more reliable connections. However, as with any new technology, it also brings new security challenges. In this project, we aim to investigate the security of private 4G/5G networks.

Specifically, we will examine the connection security between the RAN and the UE. Because telecommunications providers do not publish the inner workings of smartphone modems nor the specifics of their core network implementation, it renders it difficult to accurately study. In contrast, many RANs now follow the OpenRAN standard¹. Thus, it provides a better environment to simulate real-world network attacks.

II. BACKGROUND

A basic mobile network implementation consists of a UE, RAN, and core network. The UE is the user's device, such as a smartphone. The RAN is the base station that connects the UE to the core network. The core network is the central part of the network that handles the user's data.

¹OpenRAN is an open-source standard for RAN implementations.

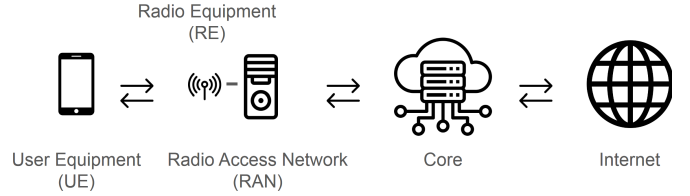


Fig. 1: Simplified Network Diagram

A. User Equipment (UE)

Each UE connected to the network contains a SIM card that stores confidential user information. Namely, it contains the International Mobile Subscriber Identity (IMSI), which is a unique identifier for the user. The IMSI typically consists of three components:

- 1) The Mobile Country Code (MCC): identifies the country of the subscriber.
- 2) The Mobile Network Code (MNC): identifies the home network operator.
- 3) The Mobile Subscriber Identification Number (MSIN): a unique identifier within the network.

When a UE connects to the network, it sends its IMSI to the RAN. The RAN then forwards the IMSI to the core network for authentication.

B. Radio Access Network (RAN)

The RAN is responsible for connecting the UE to the core network. It consists of a base station (gNodeB for 5G, eNodeB for 4G) and a software-defined radio (SDR). The SDR is used to transmit and receive radio signals between the UE and the base station.

To efficiently transmit data, the RAN uses different channels for the control plane (CP) and the user plane (UP). The CP is used for signaling messages, such as authentication and handover requests. Meanwhile, the UP is used for transmitting data for user applications.

C. Core Network

The core network is the central part of the mobile network. It consists of several microservice components, such as the Home Subscriber Server (HSS), the Mobility Management

Entity (MME), and the Serving Gateway (SGW). These function to authenticate the UE, manage mobility, and route data between the UE and the internet.

To uniquely identify each network, the core network also uses a Mobile Country Code (MCC) and Mobile Network Code (MNC). These combine to form the PLMN identifier, which is used to identify the network to which the UE is connected.

This PLMN identifier is vital to facilitate UE mobility. Once connected, when a UE physically moves into range of another RAN within the same network, the core initiates handoff. In the handoff procedure, the UE sends an NAS attach message to the new RAN containing its identity, capabilities, and security. Once authenticated, the core network transfers the UE's context to the new RAN.

D. 5G NSA Network Composition

While a 5G Standalone (SA) network is composed of exclusively 5G network components, a 5G Non-Standalone (NSA) network is composed of a 5G UE, a eNodeB, a gNodeB, and a 4G core network².

In a 5G NSA network, the UE connects to both the eNodeB and gNodeB: it uses the 4G eNodeB for control plane signaling and the 5G gNodeB for user plane data transmission. Both RANs then connect to a single core. This allows the UE to connect to the 5G RAN for fast data transmission, while still using the 4G core network for authentication and mobility management. Reusing existing 4G infrastructure allows for a faster deployment of 5G NSA networks. Consequently, all 5G networks within Switzerland are currently NSA networks.

III. OBJECTIVES

As previously mentioned, we aim to investigate the security of private 4G/5G networks. This task consists of three parts:

- 1) Set up a private 5G and 4G network to simulate a real-world scenario. This will help us understand the inner workings of the network and identify potential vulnerabilities.
- 2) Conduct an IMSI catching attack. This will involve intercepting the IMSI of a UE. By capturing the IMSI, an attacker can track the user's location and monitor their activity.
- 3) Conduct a inter-generation downgrade attack³. In this attack, we will force the UE to downgrade from UMTS to a GSM connection for voice calls. This will allow us to eavesdrop on the call and potentially intercept sensitive information.

We will attempt to perform both these attacks against the real Swisscom network.

IV. DISCLAIMER

Due to the sensitive nature of security research, we performed this project under several restrictions. The technical demon-

²To support the additional 5G functionalities, there are a few additional necessary components. However, the bulk of the 4G network remains the same.

³A downgrade attack is also sometimes known as a bidding-down attack.

strations were performed in a controlled environment with low-power antennas and amplifiers. This was done to avoid disturbing legitimate traffic and interfering with licensed bands. As a result, the range of our experiments was limited to a classroom setting. The results presented in this document are intended for educational purposes only. We do not support or encourage the use of illegal RF interference for nefarious purposes.

V. PRIVATE 4G/5G NETWORK IMPLEMENTATION

To better understand the inner workings of 4G/5G networks, we set up a simplified testbed using open-source software and off-the-shelf hardware. Because all these tools follow the 3GPP standards and are widely used in the telecommunications industry, they provide a realistic example of a commercial network.

A. Components

Our testbed consists of the following components:

- UE: Samsung Galaxy A15 5G, Android version 14 [1]⁴.
- SIM Cards: Programmable Open Cells SIM cards [2].
- Radio Equipment: bladeRF 2.0 micro xA4 [3]. This SDR is capable of transmitting and receiving signals in the 300 MHz to 3.8 GHz range.
- eNodeB: srsRAN-4G software using the workaround to support bladeRF radios [4] [5] .
- gNodeB: srsRAN-5G software [4].
- Core: Open5Gs hosted in the cloud; we can connect to the core using a DTLS 2.0 VPN tunnel [6].
- PC: Intel NUC with an Intel Core i7 processor and 16 GB of DDR4 RAM running Ubuntu 20.04 [7].

B. Setup

Mirroring the real-world 5G NSA network architecture, our testbed consists of a 5G UE, a 4G eNodeB, a 5G gNodeB, and a 4G/5G core network.

- The UE contains an OpenCell SIM card with a custom IMSI, which is also stored in the core's HSS MongoDB database.
- The eNodeB/gNodeB consist of a bladeRF SDR connected to NUCs running srsRAN-4G and srsRAN-5G software respectively.
- The core network uses a Open5GS hosted in the cloud. Our implementation uses the test network MCC "001". Each RAN connects to the core using a secure VPN tunnel from Cisco [8]⁵.

For a complete description of our 4G/5G NSA network setup please see the relevant open-source installation documentation [9] [10].

⁴While we attempted to use other UE to connect to our network, this was the only phone which could provide a consistent connection.

⁵Apart from security, we also need to connect via VPN tunnel because the MTU does not fit neatly inside IP packets.

- 1) Using a COTS smartphone, we can determine what 4G frequency bands are used by Swisscom—our target network.
- 2) Next, we employ the previously mentioned jammer to disconnect the UE from the legitimate RAN. As a result, the UE will search for a new RAN to reconnect to.
- 3) The UE locates our fake eNodeB and attempt to connect with it. Although it will not successfully connect, it still exchanges TAU attach messages containing its IMSI.
- 4) By examining the attach requests to our fake network, we can view the IMSI of every UE victim from our attack.

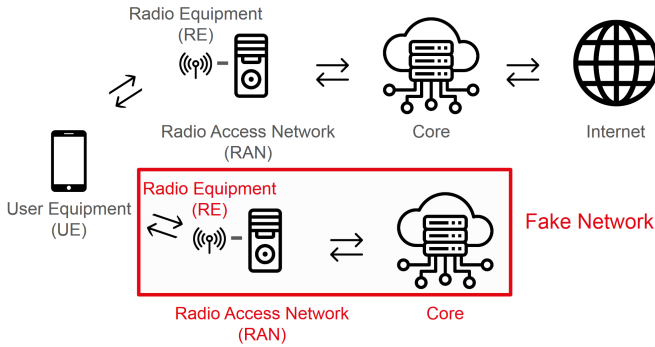


Fig. 3: Simplified IMSI Catching Attack Diagram

C. Results

With this attack, we were able to collect IMSI from all nearby UE connected to Swisscom. See Appendix C for the full results.

D. Possible Patches

This attack hinges on the UE trusting the PLMN advertised by RANs. A first step to counteracting this would be to ensure the phone verifies the integrity of RAN information before attempting to connect. Implementing PKI infrastructure and digitally signing each base station's information would ensure that UE would only attempt to connect to legitimate base stations.

VIII. INTER-GENERATION DOWNGRADE ATTACK

In an Inter-Generation Downgrade Attack, the UE is disconnected from a secure 3G UMTS voice connection and forced to downgrade its connection to insecure and spoofed 2G connection.

The UE is disconnected from a legitimate 3G UMTS networks and forced to connect to our 2G network. Once connected, outbound calls and SMS can be routed (an eavesdropped) through an attacker's SIP infrastructure, before reaching their final destination through a commercial SIP trunk.

A. Fake 2G Network

We implemented the 2G network using YateBTS with the Network in a PC configuration [12]. This consists of 2G RAN, known as a GSM, and a core network hosted on the same PC. The HLR⁷ is set to allow any UE to connect to it, and the broadcasted PLMN identity matches Swisscom's one.

B. Attack Procedure

Similarly to the IMSI Catching Attack, we employ a jammer and 2G network within close proximity to the UE victim.

- 1) Using a COTS smartphone, we can determine what 3G frequency bands are used by Swisscom—our target network.
- 2) Next, we employ the previously mentioned jammer to disconnect the UE from the legitimate 3G RAN. As a result, the UE will search for a GSM network to reconnect to.
- 3) The UE locates our fake GSM and successfully connects to it for voice calls and SMS. Although the UE is connected to our fake network for calls, it still displays that it is connected to Swisscom, using either 4G or 5G technologies.
- 4) Once the UE attempts to initiate a call, if VoLTE is not enabled, it will default to a GSM voice call over our network. We can then eavesdrop on the call. The same scenario applies for SMS messages. Another scenario would be the one where the attacker can initiate spoofed calls and SMS to the victim, impersonating contacts or trusted organizations for example.

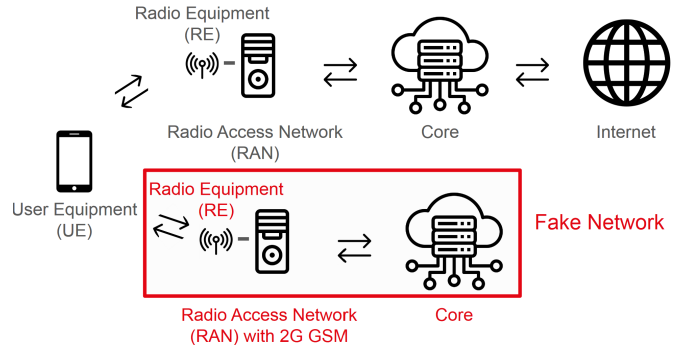


Fig. 4: Simplified Inter-Generation Downgrade Attack Diagram

C. Results

Much of the world still relies on GSM for voice calls as shown in Appendix D. As a result, they are all susceptible to this attack.

With this attack, because all GSM voice calls are unencrypted over our network, we can eavesdrop on any outgoing calls. See Appendix E for the full results.

⁷a HLR is the 2G equivalent to an HSS.

D. Possible Patches

Blocking any connections to 2G networks and making VoLTE/VoNR standard in roaming agreements would instantly render this attack useless. As such, many countries have done this; however, many have yet to.

IX. CHALLENGES

Because of our limited prior experience with 5G networks, we encountered several growing challenges during the course of this project.

A. Complicated Topics

Because 5G is still an emerging technology, it lacks many available open-source tools. Moreover, for the tools that do exist, the documentation is often poor and incomplete. Combined with the complexity of 5G 3GPP standards, it makes any amateur experimentation extremely difficult. See Appendix F for a full description of what tools we attempted to use.

B. Core Network

We initially attempted to host our core network in the cloud on a DigitalOcean VM. Traffic was routed from the RAN to the core via an OpenVPN server [13]. Using a remote core would enable us all to simultaneously connect our RANs to a single core. However, the OpenVPN solution proved far to unstable: a mix of configuration changes and high packet loss made it untenable. Thus, we switched to a private server using a Cisco DTLS 2.0 VPN tunnel. This proved far more effective.

C. gNodeB RAN

Our gNodeB setup faced several challenges. The original srsRAN_Project [4] workaround did not work. Thus, we had to switch to a fork of srsRAN_Project with bladeRF support. Once we successfully setup the network, we encountered another issue where our UE could briefly connect to the network, but would disconnect upon later attempts. After consulting with the TAs, we discovered the issue lied with our UE, and only the Samsung Galaxy A15 5G could successfully use our network. The reason why only this phone works remain unknown.

X. LEARNING OUTCOMES

Through this project, we gained valuable experience in several areas relevant to a future career in telecommunications. It would be difficult to attain a similar set of skills through traditional coursework alone.

A. Debugging Complicated Systems

The tools we used for this project were complex and poorly documented. As a result, we had to spend a significant amount of time troubleshooting builds, configurations, etc. However, this experience was invaluable as it taught us how to work with unfamiliar systems and solve problems independently.

B. Using Industry Software

To implement our private 4G/5G NSA network, we used several industry-standard tools, such as srsRAN and Open5Gs. From space-based networks to aviation, these tools are widely used in the telecommunications industry and provided us with a realistic example of a commercial network. This experience will be invaluable as we move forward in our careers.

C. Insights into being a Communications Engineer

Through this project, we gained a valuable insight into the life of a communications engineer. We learned how to set up a private 4G/5G network, conduct security attacks, and analyze the results. This project served as a perfect introduction to the field and has inspired us to pursue further study in this area.

XI. CONCLUSION

This project was an excellent introduction to understanding 5G mobile networks. It provided the opportunity to get hands-on experience working with commercial-grade 5G mobile network infrastructure. Furthermore, by investigating two security attacks, we gained invaluable insights into the authentication and encryption mechanisms used in all generations of mobile networks.

The research contained in this project proves as a baseline for future study into the intricacies of mobile networks. As with all research, there are still possible add-ons to improve the attack effectiveness. Improving the jamming method's range and discreteness are obvious first steps towards this. Furthermore, the bladeRF lack the power of a commercial radio, and thus, the attack radius is extremely small; simply using a more powerful radio would instantly improve performance ten-fold.

But, despite the countless issues and setbacks, the project was still ultimately a success and we were able to achieve our initial project objectives. This was not a trivial feat, and we are proud of the results we achieved.

REFERENCES

- [1] Samsung, "Galaxy A15 5G." [Online]. Available: <https://www.samsung.com/us/smartphones/galaxy-a15-5g/>
- [2] Open Cells Project, "Open cells Sim Card version 5." [Online]. Available: <https://open-cells.com/index.php/sim-cards/>
- [3] Nuand, "bladeRF 2.0 micro xA9 ." [Online]. Available: <https://www.nuand.com/product/bladeRF-xa9/>
- [4] Software Radio Systems, "srsran project." [Online]. Available: <https://www.srsran.com/5g>
- [5] Sorin Popescu, "Srsran with bladerf drivers." [Online]. Available: <https://github.com/hypermagik/srsRAN-5G>
- [6] Sukchan Lee, "Open 5g project." [Online]. Available: <https://open5gs.org/>
- [7] Intel, "Next Unit of Computing : NUC12WSH." [Online]. Available: <https://www.intel.com/content/dam/support/us/en/documents/intel-nuc/NUC12WS-TechProdSpec.pdf>
- [8] Cisco, "Cisco anyconnect secure mobility client." [Online]. Available: https://www.cisco.com/c/en_uk/products/security/anyconnect-secure-mobility-client/index.html
- [9] "Building open5gs from sources." [Online]. Available: <https://open5gs.org/open5gs/docs/guide/02-building-open5gs-from-sources/>
- [10] "srsran project documentation." [Online]. Available: <https://docs.srsran.com/projects/project/en/latest/index.html>

- [11] jhonnybonny, "Clever jam." [Online]. Available: <https://github.com/jhonnybonny/CleverJAM>
- [12] S.C. NullTeam Impex SRL, "Yatebts." [Online]. Available: <https://yatebts.com/documentation/radio-access-network-documentation/radio-network-troubleshooting/>
- [13] OpenVPN Project, "Openvpn." [Online]. Available: <https://openvpn.net>
- [14] SRS, "Srsran 4g." [Online]. Available: https://github.com/srsran/srsRAN_4G
- [15] "Kamailio." [Online]. Available: <https://www.kamailio.org/>
- [16] Nuand, "BT-100 Bias-tee Power Amplifier." [Online]. Available: <https://www.nuand.com/product/bt-100/>
- [17] Yaxiong Xie, "Ng-scope." [Online]. Available: <https://github.com/YaxiongXiePrinceton/NG-Scope>
- [18] H. Wan and K. Jamieson, "Evolving Mobile Cloud Gaming with 5G Standalone Network Telemetry." arXiv, Feb. 2024, arXiv:2402.04454 [cs]. [Online]. Available: <http://arxiv.org/abs/2402.04454>
- [19] System Security and Applied Cryptography Laboratory (SysSec Lab), "Ng-scope." [Online]. Available: <https://github.com/SysSec-KAIST/LTESniffer>
- [20] T. D. Hoang, C. Park, M. Son, T. Oh, S. Bae, J. Ahn, B. Oh, and Y. Kim, "LTEsniffer: An Open-source LTE Downlink/Uplink Eavesdropper," in *Proceedings of the 16th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, ser. WiSec '23. New York, NY, USA: Association for Computing Machinery, June 2023, pp. 43–48. [Online]. Available: <https://doi.org/10.1145/3558482.3590196>
- [21] Fast Analysis of LTE Control channels, "Falcon." [Online]. Available: <https://github.com/falkenber9/falcon>
- [22] R. Falkenberg and C. Wietfeld, "FALCON: An accurate real-time monitor for client-based mobile network data analytics," in *2019 IEEE Global Communications Conference (GLOBECOM)*. Waikoloa, Hawaii, USA: IEEE, Dec. 2019. [Online]. Available: <https://arxiv.org/abs/1907.10110>
- [23] 5G Media Action Group (MAG), "Mbms modem." [Online]. Available: <https://github.com/5G-MAG/rt-mbms-modem>
- [24] N. Ludant, P. Robyns, and G. Noubir, "From 5G Sniffing to Harvesting Leakages of Privacy-Preserving Messengers," in *2023 IEEE Symposium on Security and Privacy (SP)*, May 2023.

XII. APPENDIX

A. 4G/5G NSA Network Implementation

We have implemented a 4G network with srsRAN-4G [14]. We used the same core for both 5G and 4G, so the name "NSA". Two options were available for the core setup : install the core with a package manager, or build from source. As our project demanded code modifications on the core, we built it from source. We then had to configure it, most of the setup is done in the sample.yaml configuration file. In this file, every MCC and MNC field should correspond to the one we will choose with ran. For testing purpose we went with "001" for the MCC which is recognized as the code to use for test networks. We then arbitrarily choose "01" regarding the MCC. For a more complete description of the setup procedure, open5Gs has a proper installation documentation [9]. The Home Subscriber Service (HSS) based MySQL database was replaced with an unified one based on MySQL. It was accessible through FoHSS web interface. For the IP Multimedia Subsystem (IMS) we tested Kamailio, an open source SIP server [15]. It allowed us to place calls and send SMS text message. The RAN was build using the open source software srsRAN-Project [4]. This software makes it easy to deploys gNodeBs. Once the software is correctly installed, we had to modify the configuration file that will run on the gNodeB to meet the core network parameters. We followed part of the provided documentation [10]. For setting up the ran the main parameters are:

- amf : this field contains the IP address of the amf part of the core network and the binding address, which is the address to which the gNodeB binds to.
- device_driver: corresponds to the driver of the software defined radio
- plmn: this field should correspond to the MCC and MNC of the core network. For our testing purposes it was set to MCC "001" and MNC "01"

Once the configuration file has been correctly modified, it only requires to run a single command to deploy the RAN. If the core is running and all parameters are correct, the network is ready.

B. CleverJam Details

The tool we used for jamming [11] was a very simple python script based on the OsmoSDR library. It used uniformly distributed noise to jam bands with equals power. We first tried to jam the Downlink channel between the BladeRF and the UE. Due to intensive usual interference and traffic this method was not constant. Also we have to take into account that the UE will try to connect others technologies like 2G or 3G or even change bands inside one generation. In order to prevent we introduced band hopping jamming to jam a list of pre measured bands. Another fact to take into account was the relatively low power output of the BladeRF which introduce difficulties to jam real life GnodeB, with far more transmitting power. We tried the use of a Faraday cage to try to reduce external interference with more powerful antennas provided by the laboratory. The solution was finally to use an bias-tee power amplifier [16] between the antenna and the BladeRF. The signal was then powerful enough to jam a few meters around the antenna. We did not use two transmitting antenna of BladeRF because it is no capable of transmitting on both of its TX antenna at the same time (2xMIMO). During testing we obtained constant jamming at the bottom floor while at the third floor we did not have any jamming effect. We think that was due to the presence of a more powerful GnodeB next to where we tested and the proximity of the air interface.

C. IMSI Catching Attack Results

Using a jammer we are able to successfully disconnect UEs from a real BS. We were able to collect IMSI from around us at a range of approximately 3 to 5 meters (due to the power of the jammer). All was monitored using the Open5G log console. Here is an example output of the console with a censored captured IMSI :

```
2024/06/07 15:26:12:750 [mme]
DEBUG : IMSI [22805*****]
NAS-EPS Type [1]
(.../src/mme/mme-context.c:3567)
```

During a 15 minutes testing we were able to catch up to 5 IMSI in a medium crowded area.

